

Motivating Claims

- Philosophical accounts of the structure and content of theories (of the general “Theory T ” type) ignore how theories are actually *applied*: the picture of theories as licensing $T \rightarrow E$ deductions is analogous to Brigham Young’s instruction to “go straight south.” Analysis of the strategies scientists actually employ requires applied mathematics rather than logic, and these strategies typically involve “physics avoidance” (forms of “deductive evasion,” p. 52).
- This descriptively false account of theories leads to philosophical mistakes, in part by neglecting the overall “explanatory landscape” in which concepts (causation, possibility, law, force, ...) function. Ramifications for central topics in philosophy of science: interpretation of theories, causation, laws and modalities, determinism, realism. (Alongside Wilson’s broader critique of the “classical theory of concepts” in *Wandering Significance*, and the recent *Imitation of Rigor*.)

Theory T Thinking

Illustration: “Standard Account” for Classical Mechanics

- *State* of the system: list of all the information needed to characterize the system. Generate (minimal) abstract space, equipped with (at least) topology Γ : $p \in \Gamma$ corresponds to specific state.
- *Dynamics*: a rule relating a point $g \in \Gamma$, the “initial state” at a time t_0 , to other points in the state space representing the system at earlier and later times (assuming it “evolves undisturbed”). More precisely $D_{\Delta t} : \Gamma \rightarrow \Gamma$, sends initial point g to g' which represents the state at time Δt away. (For deterministic dynamics there is unique g' for all Δt .)
- *Observables*: quantity that has a specified value for all states of the system; abstractly an observable A is a map: $f_A : \Gamma \rightarrow \mathbb{R}$. For a given state $g \in \Gamma$, the “value of A ” in that state is $f_A(g)$.
- *Semantics*: we then have a natural way of reading off the truth values of various propositions regarding the state of the system: “Observable A has a value falling in the interval I ” will be true iff, given that the state of the system is g , $f_A(g) \in I$. For any well-formulated proposition P , there is a subset Γ_P (possibly the empty set) which includes all the states such that P is true. (This can be generalized to describe propositions that assign numerical values to properties.)
- *Evolutionary Modeling*: phase space Γ reflects all possible states of the system at a time ($p \in \Gamma$ a full characterization of its properties); laws pick out trajectories $D_{\Delta t} : \Gamma \rightarrow \Gamma$; “contingencies” (initial conditions) reflected in choice of trajectory. The view treats laws as capturing the essential dynamical features of the system, relatively uninterested in the details of application in particular contexts (all reflected in initial / boundary conditions).

To what extent does this simple picture of the basic structure of physical theories reflect real physics? Or foster philosophical misconceptions?

Railton’s “Ideal Explanatory Text”

- Ideal text includes “full state specifications” at finite time-steps Δt , connected through logical deductions $t_i \rightarrow t_{i+1}$. Generalization of Hempel’s DN model to include probabilistic laws.
- Explanation requires *condensing* this text—determining what is relevant for particular explanatory purposes.
- Wilson criticizes Railton for several reasons: (i) it is a mistake to equate “laws” with the differential equations that govern an “evolutionary flow”; (ii) the conception of “causal process” is inadequate (closer to common-sense finite differences than to differential operators); (iii) one must use applied-math tools to give an account of “physics avoidance” strategies in place of Railton’s appeal to “condensations” or to vague criteria of relevance.
- “*Man proposes, Nature disposes*”: the interplay between theory and evidence reduces, on the Theory T picture, to checking predictions.

Wilson’s pragmatic account: a much richer interplay—theory guiding inquiry, design, practical interventions, etc. (a line of criticism developed in other chapters). Furthermore, success does not flow upward from “fundamental physics”; the relationship between regimes where applications have been successful and “more fundamental” physics is typically unclear.

Background: Partial Differential Equations (see, e.g., Olver 2014)

Consider a general, homogeneous linear second-order PDE for a function $u(x, t)$, where $u_{tt} = \frac{\partial^2 u}{\partial t^2}$ (etc.):

$$L[u] = Au_{tt} + Bu_{tx} + Cu_{xx} + Du_t + Eu_x + Fu = 0 \quad (1)$$

The classification of PDEs depends on the *discriminant*, $\Delta = B^2 - 4AC$. Four types:

1. *Hyperbolic*: $\Delta > 0$. Wave equation, $u_{tt} - u_{xx} = 0$.
2. *Parabolic*: $\Delta = 0$ and $A^2 + B^2 + C^2 \neq 0$. Heat equation, $u_t - u_{xx} = 0$.
3. *Elliptic*: $\Delta < 0$. Laplace’s equation, $u_{xx} + u_{tt} = 0$.
4. *Singular*: $A = B = C = 0$ (...Examples?)

This classification generalizes straightforwardly in two respects: to functions of several variables; and to coefficients that are functions, $A(x, t), B(x, t), \dots$, in which case the type of the PDE may vary over the domain. (The names derive from an analogy with the quadratic equation, whose solutions are plane curves of the corresponding types.)

The study of PDEs divides into subfields following this typology, due to fundamental differences in the character of solutions, solution techniques, and so on. Wilson’s comments regarding shifting “explanatory landscapes” track these contrasts. (See also Smith (2000) for a discussion of contrasting notions of causality appropriate in each setting.)

Hyperbolic Equations

Features exemplified by the wave equation (where c is the wave speed): $u_{tt} - c^2 u_{xx} = 0$.

- *Initial value problem*: given specified initial values ($u(x, 0) = f(x)$ and its first derivative $u_t(x, 0)$), find a C^2 solution $u(x, t)$ to the equation.

- For the 1D wave equation there are “right-moving” solutions, $p(x - ct)$, and “left-moving” solutions $q(x + ct)$. General solution: $u(x, t) = p(x - ct) + q(x + ct)$.
- *Causal structure*: two characteristics (right and left) from any point on the line mark the domain of influence (the solution within this domain depends on the initial data at the point); in $3 + 1$ dimensions this yields a cone-shape (the light cone for electromagnetic waves). Independent perturbations at separated points cannot communicate faster than the characteristic speed—this is what makes the equations a natural vehicle for “causal process.”
- *Character of solutions*: non-analytic functions (see Hadamard quote on p. 74); more general hyperbolic equations allow discontinuities (“shock waves”).
- *Solution techniques*: “marching method,” typically of evolutionary character (reflecting causal structure).

Elliptic Equations

Features exemplified by Laplace’s equation, where t is often replaced with y to acknowledge that the equation describes an equilibrium situation: $u_{xx} + u_{yy} = 0$.

- *Boundary value problem*: seek a solution $u(x, y)$ over some bounded region Ω with boundary values for u specified on $\partial\Omega$ (Dirichlet: fix u on the boundary; Neumann: fix the normal derivative).
- *Solutions*: (linear sums of) harmonic functions, representing the static or equilibrium state of a system. (No immediate sense of “dynamical evolution.”) Analytic functions “secretly embody a reproducibility property”: specifying the function on an arbitrary small neighborhood uniquely determines it over Ω (analytic continuation). This is exactly the *wrong* property for a causal process—it would mean that hammering on one end of a beam fully determines the behavior at the other end, with no signal travelling between.
- *Solution techniques*: “shooting method”—iterate guesses to find a solution consistent with the boundary conditions on all sides simultaneously.

Shifting Explanatory Landscapes

Two main types of “physics avoidance” strategy (see also Chapters 4, 6):

- Shifting from hyperbolic to elliptic PDEs—an evolutionary problem to an equilibrium problem. (Often justified by separation of time scales: neglect *fast* evolutionary processes and assume relaxation back to equilibrium.)
- Eliminating variables by introducing constraints. (Often justified by separation of length scales, with constraints treating some configurations as rigid; constraint “forces” enter as Lagrange multipliers.)

But the idea has broader implications:

I sometimes invoke the notion in applicational circumstances where we presently lack any clear conception of what we're actually avoiding. Wary readers should also become alerted to the fact that elementary physics primers often switch from modeling strategy *A* to modeling strategy *B* with an offhanded remark to the effect that it is “obvious” that *B* is fully supported as an “approximation” to *A*. Like the emperor's new clothes, such appeals often point to a justificatory fabric that simply isn't there. (p. 78; elsewhere this is called the “lousy encyclopedia” phenomenon, an Escher-staircase structure.)

The Loaded Strut

Picture a slender vertical column—a strut—loaded with a weight W on top. The column may flex side-to-side as $x(y, t)$, where y runs along its length and t is time. Modeled as a flexible band, with damping, we obtain:

$$\rho \partial_{tt}u - EI \partial_{yy}u + Wu + \kappa \partial_t u = 0, \quad (2)$$

where ρ = density, EI = “flexural rigidity” (a measure of how stiff the column is), κ = frictional damping, W = external load. This is hyperbolic in (y, t) . Two natural questions:

- *Problem 1: How much weight can my strut hold before it buckles?*

Civil engineers do not want to track the wave dynamics inside a column; they want to know whether a static column under a given load will collapse. Drop the time-dependent (first and third) terms to obtain *Euler's reduced equation*, an elliptic equation that describes equilibrium states and from which the “critical load” for buckling can be calculated.

But this is a substantive change of explanatory landscape: the reduced equation no longer describes “time evolution” of the strut at all. There is no longer any *internal* time variable for the system, although we can reintroduce an *external* time (“manipulation time,” p. 75) by slowly increasing the load and tracking the sequence of equilibria.

More generally (e.g. steady-state fluid flow in §(i), monochromatic light scattering in Appendix 1): Wilson diagnoses various confusions that arise when concepts like “cause” and “time,” appropriate to evolutionary models, are applied within this new landscape.

- *Problem 2: How does a more detailed model (describing the propagation of stresses through the strut in response to the load) relate to the simple equilibrium picture?*

Appealing picture: the strut undergoes rapid relaxation to equilibrium, with internal time-scales very short compared to the external manipulation time-scale. But “our presumption that the success of Euler's equilibrium based strategy can be founded upon an underlying evolutionary modeling of the sort suggested should be regarded as an *aspirational hope*, not an established verity” (p. 77).

The damping term $\kappa \partial_t u$ is a placeholder for processes (vibrations dissipated in the clamps, in the surrounding air, etc.) whose accurate evolutionary modeling would itself be a major undertaking. Even with the damping term included, there is no guarantee that the system asymptotes to the equilibria *A*, *B*, or *C* that the reduced equation describes.

More generally: successful modeling often presupposes substantive assumptions about underlying physics that have not been independently verified. Wilson's term for this is *mimicry*: a pattern *X* (Euler's reduced equation) presumed to piggyback on an evolutionary modeling *Y* may in fact be justified, if at all, by quite different underlying factors *Z* (e.g. quantum-mechanical sources of rigidity).

Constraints

The second main mode of physics avoidance is the introduction of *constraints* that reduce the number of independent variables, exploiting separation of length scales rather than time scales.

- *Setup*. Treat some component of the system as *rigid* (or some fluid as *incompressible*, or a bead as *confined to a wire*), rather than modeling its internal degrees of freedom. The constraint enters as an additional, non-evolutionary equation (in the rigid-bar case, an algebraic equation fixing distances between marked points).
- *Two length scales*. At a fine scale ΔL , a bead in fact wobbles about a wire; an iron bar in fact deforms. But at a coarser scale ΔL^* the constraint holds. “Rigidity” is a ΔL^* -level fact about the system.
- *The Lagrangian trick*. If we try to model the system at the fine scale ΔL using $F = ma$ alone, we lack force laws for the constraints—the problem is *under-constrained*. Lagrange’s resolution: introduce auxiliary “force centers” inside the rigid components, with strengths λ_i (Lagrange multipliers). A supplementary principle (least work / virtual work) fixes the λ_i at the values needed to maintain the constraint at ΔL^* .
- *Analogy with combination of observations*. Wilson draws a structural parallel with the astronomical combination-of-observations problem: a few orbital parameters are fit to many noisy measurements via least squares. The λ_i play the same structural role as the error terms ϵ_i , and the variational principles play the same role as the least squares fit.
- *Macroscopic data, microscopic forces*. The strengths of the constraint forces are determined by what they must induce *at the macroscopic level* (rigidity). In effect, Lagrangian methods import macroscopic measurement-based information into a microscopic-looking formalism. Wilson calls this a *mixed-level explanation*—neither pure bottom-up nor pure top-down.
- *Conceptual fallout*. “Force” migrates in meaning: λ_i -type “forces” do not satisfy Newton’s third law in the standard way, and what underwrites the rigidity of an iron bar—if anything classical does—may not be available at the same level as the active forces in the model. Wilson takes this referent-migration to be a source of the lingering philosophical sense that there is something “fishy” about force.

Concluding Discussion

- *Replacement or reform?* To what extent has Wilson offered a *replacement* for the Theory T view, as opposed to a new agenda for items it neglects? How much of the standard picture survives?
- *Axiomatization*. What is the appropriate role of axiomatization and other formal approaches? (See also *Imitation of Rigor*: there is no reason to expect that a single axiomatization will suffice; criticisms of the scope of axiomatizations.)
- *Realism*. In what sense is Wilson a “scientific realist at heart” (p. 79)? (See also the recent *PPR* book symposium.) How does this kind of realism compare with van Fraassen’s empiricism, and with the position implicit in Smith & Seth?
- *Scope: applications or fundamentals?* Do shifting explanatory landscapes arise only in applications, or already in the basic equations of physics?

A case could be made that, at least on a fundamental level, all the “partial differential equations of physics” are hyperbolic — that, e.g., elliptic and parabolic systems arise in all cases as mere approximations of hyperbolic systems. Thus, Poisson’s equation for the electric potential is just a facet of a hyperbolic system, Maxwell’s equations. (Geroch 2017, p. 21)

- *Forward look: EFT.* Wilson’s emphasis on cross-scale modeling and limited inter-scale justification anticipates effective field theory methods. How does the EFT framework relate to Wilson’s physics-avoidance themes?

References

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